Notes: Andersen, Campbell, Nielsen, and Ramadorai (AER, 2020)

Sources of Inaction in Household Finance: Evidence from the Danish Mortgage Market

Contents

1 Big picture

- **Question.** Why do households delay mortgage refinancing even when the interest savings are large, and can we separate different sources of inaction?
- **Setting.** The paper studies Danish fixed-rate mortgage borrowers from 2009–2017. Denmark is ideal because fixed-rate mortgages can be refinanced without interest-rate penalties, and non-cash-out refinancing is not constrained by negative home equity, deteriorated credit quality, or upfront liquidity needs.
- **Core empirical challenge.** A household may fail to refinance because it has a high psychological cost of taking action, because it rarely pays attention to refinancing opportunities, or because the observed financial incentive is below the true optimal threshold. These explanations can look similar in a simple cross-section.
- **Main conceptual distinction.**
  - **State-dependent inaction:** households refinance only when the financial incentive is strong enough to overcome monetary and psychological fixed costs.
  - **Time-dependent inaction:** households consider refinancing only intermittently, so the probability of action is low even when incentives are strong.
- **Main method.** The authors build an empirical mixture model that combines a state-dependent refinancing threshold with a Calvo-style probability that a household is "asleep" and does not respond to incentives in a given period.
- **Main data advantage.** Danish administrative data observe both refinancers and non-refinancers, with detailed demographics, income, wealth, education, and mortgage terms. This makes it possible to compare who leaves money on the table.
- **Main empirical result.** Middle-aged and wealthy households behave as if they have high psychological refinancing costs. Older, poorer, and less-educated households instead refinance with low probability regardless of incentives, consistent with high information-gathering or attention costs.
- **Main policy result.** Policies that "wake up" inattentive households can be more powerful than simply subsidizing refinancing fees, especially for older and lower-income households.
- **Economic takeaway.** Mortgage refinancing failures are not only about standard fixed costs or borrowing constraints. Attention and information frictions weaken the mortgage refinancing channel of monetary policy, because lower rates help fixed-rate borrowers only when they actually refinance.

2 Data and measurement (key objects)

2.1 Danish mortgage system

- **Mortgage design.** Denmark has long-term fixed-rate mortgages that are prepayable without penalties, similar to the United States.
- **Funding structure.** Danish mortgages are funded through covered bonds issued by mortgage banks. Mortgage banks intermediate between investors and borrowers, but market prices of the covered bonds determine mortgage rates.
- **Key refinancing feature.** Borrowers can refinance without cashing out even if home equity is negative or credit quality has deteriorated.
- **Why this matters.** In the United States, refinancing failures may reflect credit-score, loan-to-value, or origination-market constraints. In Denmark, those barriers are largely absent for non-cash-out refinancings, so delayed refinancing is much more directly informative about household inaction.
- **Fees.** Refinancing costs increase with mortgage size, but they do not vary with the level of interest rates. Refinancing costs can also be rolled into the new mortgage balance.
- **Old coupon versus new yield.** Because Danish mortgage bonds are issued with discrete coupons, the refinancing incentive is based on the coupon rate on the old mortgage bond relative to the yield on a new mortgage, net of an optimal refinancing threshold.
- **Cash-out.** Cash-out refinancing does require positive home equity and good credit standing. The paper includes these refinancings in the main results but shows robustness to excluding them.

2.2 Administrative data sources

- **Mortgage data.** Mortgage-level data come from Danmarks Nationalbank, based on information from all mortgage banks through Danish mortgage-banking associations.
- **Demographics.** The Danish Civil Registration System gives household identifiers, age, gender, marital status, spouse and children information, and household composition.
- **Income and wealth.** The Danish Tax Authority provides third-party-reported income and financial information. This is unusually reliable because much taxation occurs at source.
- **Education.** The Ministry of Education provides the highest completed education and professional qualifications.
- **Mortgage rates.** Current mortgage yields are constructed from mortgage bond yields, grouped by maturity bands.

2.3 Sample construction

- The sample covers adult Danish households from **2009–2017**.
- A household is one or two adults at the same postal address.
- The analysis focuses on households with a **single fixed-rate mortgage** that can be tracked over time.
- For the 2009–2010 consecutive-year period, the authors begin with 960,159 households with any mortgage, then 641,786 households with a single mortgage, then 330,350 households with a single fixed-rate mortgage.
- The final analysis sample contains **2,376,815 household-year observations**, expanded to **9,351,183 household-quarter observations**.
- The paper observes **378,421 refinancings** over the sample period, implying an average refinancing rate of about **4 percent per quarter**.
- Of these refinancings, **113,333** are from fixed-rate to adjustable-rate mortgages and **265,088** are from fixed-rate to fixed-rate mortgages or capped adjustable-rate mortgages.

2.4 Key descriptive objects

- The average mortgage has principal remaining of **0.983 million DKK**, roughly US$147,000.
- The average remaining maturity is **23.226 years**.
- The average loan-to-value ratio is **0.600**.
- The refinancing rate is **0.040** per quarter.
- About **70.1 percent** of refinancings are fixed-rate to fixed-rate.
- About **56.3 percent** of household-quarter observations have negative refinancing incentives relative to the ADL benchmark, and only **1.3 percent** of those refinance.
- About **43.7 percent** have positive refinancing incentives, but only **7.6 percent** of those refinance in that quarter.
- At the household level, about **49.8 percent** never refinance, **40.2 percent** refinance once, **8.9 percent** refinance twice, and **1.1 percent** refinance three or more times.

2.5 Unobservables and timing

- **Observed state variables.** The model uses mortgage characteristics, current refinancing incentives, household demographics, education, income, financial wealth, housing wealth, region, and time effects.

- **Unobserved psychological costs.** The model allows refinancing costs to include a psychological component beyond direct monetary fees.
- **Time-dependent inaction.** Each period, a household may be "asleep," meaning it does not evaluate or act on the refinancing opportunity.
- **Stochastic choice shock.** Conditional on being awake, refinancing is probabilistic through a logit shock.
- **Timing.** In each household-quarter, the mortgage terms and current rates imply a financial incentive; if the household is awake, it responds according to the state-dependent refinancing rule; if asleep, it does not refinance.

3 Models and empirical specifications (all equations + interpretation)

3.1 Refinancing incentive

The household-level refinancing incentive is

$$I_{it} = \left(C_{it}^{old} - Y_{it}^{new}\right) - O_{it}. \tag{1}$$

Here:

- C_{it}^{old} is the coupon rate on the old mortgage bond.
- Y_{it}^{new} is the yield on a new mortgage.
- O_{it} is the optimal refinancing threshold.

Interpretation. The household should refinance when the coupon saving exceeds the threshold. The threshold is positive because refinancing involves fixed costs and because waiting has option value: rates might fall further in the future.

3.2 ADL optimal refinancing threshold

The paper uses the Agarwal–Driscoll–Laibson (ADL) closed-form approximation as the benchmark rational threshold:

$$O_{it} = \frac{1}{\psi_{it}} \left[\varphi_{it} + W\left(-\exp(-\varphi_{it})\right)\right], \tag{2}$$

where

$$\psi_{it} = \frac{\sqrt{2(\rho + \lambda_{it})}}{\sigma}, \tag{3}$$

and

$$\varphi_{it} = 1 + \psi_{it}(\rho + \lambda_{it})\frac{\kappa(m_{it})}{m_{it}(1-\tau)}. \tag{4}$$

The monetary refinancing cost is approximated by

$$\kappa(m_{it}) = 3000 + \max\{0.002m_{it}, 4000\} + 0.001m_{it}, \tag{5}$$

with winsorization near the ninety-ninth percentile, just below 10,000 DKK.

The effective mortgage termination rate is

$$\lambda_{it} = \mu_{it} + \frac{Y_{it}^{old}}{\exp(Y_{it}^{old}T_{it}) - 1} + \pi_t. \tag{6}$$

Here:

- ρ is the discount rate.
- σ is the volatility of interest-rate changes.
- τ is the tax rate for mortgage-interest deductions.
- m_{it} is mortgage size.
- μ_{it} is the exogenous mortgage termination hazard.
- T_{it} is remaining maturity.
- π_t is inflation.

Interpretation. Larger monetary costs, smaller principal, and shorter remaining maturity raise the threshold. For a small or old mortgage, the same DKK cost is large relative to the interest savings, so the household needs a bigger rate reduction to make refinancing worthwhile.

3.3 Why a pure state-dependent model fails in panel data

A pure state-dependent model says that each household has a fixed refinancing threshold. Once the incentive crosses that threshold, the household refinances.

This model can explain almost anything in one cross-section because unobserved thresholds can rationalize the observed refinancing rate at each incentive. But panel data impose stronger restrictions:

- A household should not refinance today at an incentive lower than the maximum incentive it previously ignored.
- If a household has refinanced once, it should refinance again whenever the same incentive is reached.

The data strongly reject these implications:

- About **35 percent** of first refinancings occur at incentives below the best earlier incentive the household had already experienced.
- Among repeat refinancers, the difference between the incentive at the second refinancing and the first refinancing has a standard deviation of about **327 basis points**, far from the point mass at zero implied by a fixed-threshold model.

Interpretation. Households do not behave as if they continuously monitor rates and act once a fixed idiosyncratic threshold is reached. There must be time variation in whether they consider refinancing or in their responsiveness to incentives.

3.4 State-dependent refinancing with psychological costs

Conditional on being awake, the household refinances with probability

$$p_{it}(y_{it} = 1 \mid z_{it}; \phi, \theta_i, \beta) = \Pr\left(\exp(\beta) I^*(z_{it}; \phi, \theta_i) + \varepsilon_{it} > 0\right), \tag{7}$$

where ε_{it} is standard logistic.

The key modification to the rational benchmark is that the refinancing cost becomes

$$\kappa^*(m_{it}, z_{it}; \phi) = \kappa(m_{it}) + \exp(\phi' z_{it} + \theta_i), \tag{8}$$

with

$$\theta_i \sim N(0, \sigma_\theta^2). \tag{9}$$

Interpretation. The direct monetary refinancing fee is observable, but households may also face psychological or time costs. These costs shift the refinancing threshold upward. A household with a high psychological cost will not refinance until interest savings are large.

3.5 Time-dependent inaction: the asleep probability

The paper then adds time-dependent inaction. A household is asleep in period t with probability

$$w_{it}(\chi) = \frac{\exp(\chi' z_{it})}{1 + \exp(\chi' z_{it})}. \tag{10}$$

The household is awake with probability $1 - w_{it}(\chi)$.

The period likelihood is a finite mixture:

$$\mathcal{L}_{it}(\chi, \phi, \theta_i, \beta) = w_{it}(\chi)\mathcal{L}_{it}^{asleep}(\phi, \theta_i, \beta) + (1 - w_{it}(\chi))\, \mathcal{L}_{it}^{awake}(\phi, \theta_i, \beta). \tag{11}$$

The sample log likelihood is

$$\log \mathcal{L}(\chi, \phi, \theta_i, \beta) = \sum_t \sum_i \log \mathcal{L}_{it}(\chi, \phi, \theta_i, \beta). \tag{12}$$

Interpretation. The model separates two different margins. The horizontal location of the refinancing-probability curve captures the psychological cost threshold. The height of the curve captures the probability that households are awake and able to respond to incentives at all.

3.6 Baseline model selection

The paper estimates a sequence of nested models.

- A simple logit with only incentives fits poorly.
- A pure psychological-cost model implies implausibly high psychological costs.
- A pure time-dependent model improves fit but misses state-dependent threshold behavior.
- The preferred model includes current-quarter effects, mortgage-age effects, and demographics in the asleep probability, while also allowing demographics to shift psychological costs.

The baseline is **Model 10** in Table 2:

- $\beta = 0.820$.
- average psychological-cost parameter $\phi_m = 2.547$.
- asleep-probability parameter $\chi = 3.059$.
- pseudo-$R^2 = 0.069$ relative to the restricted benchmark.
- A random-coefficients extension raises pseudo-R^2 only from 0.069 to 0.070, so the authors keep the simpler baseline.

For a reference household, the baseline implies psychological refinancing costs of about **DKK 12,768**, roughly **US$1,914**, and an asleep probability of about **96 percent** in the reference quarter.

Interpretation. The model does not just say that households are slow. It decomposes slowness into a threshold component and an attention/consideration component.

3.7 Policy simulation design

The paper uses the model to simulate an interest-rate decline that makes **90 percent** of households have a positive incentive relative to the ADL threshold.

The main scenarios are:

- fully rational refinancing at the ADL threshold;
- a pure state-dependent model with psychological costs;
- a pure time-dependent model;
- the baseline model with both frictions;
- a refinancing-fee rebate policy;
- a wake-up policy that cuts asleep probabilities in half;
- a combined rebate plus wake-up policy.

Interpretation. The simulations show how different frictions affect monetary transmission. A fee rebate lowers the state-dependent threshold, but a wake-up policy raises the probability that households notice and act on the opportunity.

4 Identification and empirical credibility

4.1 What is hard to identify

- Failure to refinance can reflect many things: rational option value, monetary fixed costs, psychological costs, inattentiveness, poor information, or standard borrowing constraints.
- In a single cross-section, a distribution of unobserved thresholds can explain almost any observed pattern of refinancing.
- The paper therefore needs both a clean institutional setting and panel restrictions to separate mechanisms.

4.2 Why Denmark is useful for identification

- Non-cash-out refinancing is not blocked by negative home equity, bad current credit quality, or a lack of liquid assets to pay fees upfront.

- Mortgage products are homogeneous, rates are market-determined, and banks charge similar fees.
- The data observe non-refinancers as well as refinancers, so the authors can compute missed refinancing opportunities for the whole population.
- Household administrative data allow the authors to measure age, education, income, wealth, family structure, region, financial literacy, and life events over time.

Interpretation. The setting greatly reduces the concern that observed inaction is really hidden borrowing constraint. The remaining inaction is therefore more plausibly behavioral, informational, or attention-related.

4.3 What identifies psychological costs

- Psychological costs are identified from how strongly refinancing probability responds to the financial incentive.
- A household with high psychological costs will rarely refinance at low positive incentives but will tend to refinance once incentives become strong enough.
- In the model, these costs shift the refinancing-probability curve horizontally by raising the threshold.
- Cross-sectional variation in demographics identifies how age, education, income, financial wealth, and housing wealth affect this threshold.

4.4 What identifies time-dependent inaction

- Time-dependent inaction is identified from low refinancing probabilities that are relatively insensitive to the level of incentives.
- If a household is asleep, stronger incentives do not immediately produce refinancing.
- Quarter fixed effects and mortgage-age effects identify common and mortgage-life-cycle variation in attention or information acquisition.
- Panel evidence is critical: households often refinance after previously ignoring stronger incentives, which is hard to reconcile with a fixed-threshold model.

4.5 Robustness and threats

- **Alternative samples.** Results are robust to excluding cash-out and maturity-extension refinancings, excluding fixed-rate-to-adjustable-rate refinancings, and restricting the sample to larger, longer-maturity mortgages.

- **Alternative threshold parameters.** Reducing interest-rate volatility by 50 percent lowers the ADL threshold by about 20 basis points; reducing the discount rate by 50 percent lowers it by less than 10 basis points. These changes do not alter the main conclusions.
- **Alternative optimal refinancing model.** Using Chen–Ling thresholds instead of ADL thresholds gives similar cross-sectional patterns. The ADL and Chen–Ling thresholds have a correlation of about **0.85**.
- **Financial-advice benchmark.** Danish financial advisers give recommendations broadly consistent with the shape of the ADL threshold.
- **Prompt refinancers.** Households in the first few percent of their loan cohort to refinance do so at small incentives relative to the ADL threshold: about **24**, **49**, and **64** basis points using 2.5, 5, and 10 percent prompt-refinancer cutoffs.

4.6 Relation to the literature

- **Mortgage refinancing.** The paper contributes to work on slow refinancing and the mortgage channel of monetary policy, especially by using an institutional setting without many US-style refinancing constraints.
- **Household finance.** It connects slow refinancing to broader evidence that households make costly financial mistakes, especially when financial capability is low.
- **Inaction models.** The paper imports ideas from state-dependent and time-dependent price-setting models into household finance.
- **Behavioral economics.** The model provides an empirical way to estimate the prevalence of attention or consideration frictions when information gathering itself is not directly observed.

5 Tables and figures

5.1 Figure 1 and Table 1: ADL threshold and sample facts

Read:

- Figure 1 shows that the ADL threshold is increasing and concave in fixed refinancing costs.
- The threshold is higher for smaller mortgages and somewhat higher for shorter remaining maturities.
- Table 1 shows that the average mortgage is large and long-lived, with principal around 0.983 million DKK and 23.226 years remaining.

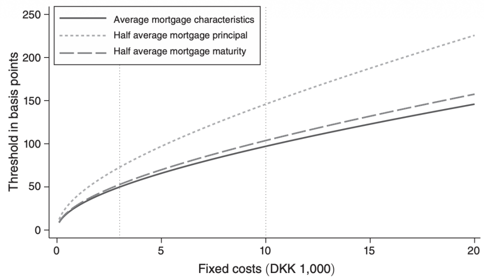


FIGURE 1. ADL THRESHOLD AS A FUNCTION OF FIXED COSTS

Notes: This figure plots the ADL threshold level in basis points associated with each fixed cost in DKK on the *x*-axis. The solid line in the plot shows this mapping when the ADL threshold is computed using the mean estimated mortgage termination probability, the mean remaining mortgage principal, the mean remaining horizon on the mortgage, the mean interest rate on the mortgage, and the mean inflation rate. The two dashed lines in the plot show this mapping for (i) a smaller mortgage that is one-half of the mean principal, and (ii) a shorter duration mortgage with one-half of the mean remaining horizon to maturity.

TABLE 1—CHARACTERISTICS OF DANISH FIXED RATE MORTGAGES

Panel A. Household quarter observations			
Observations		9,351,183	
Principal remaining (million DKK)		0.983	
Years remaining on mortgage		23.226	
Loan-to-value (LTV) ratio		0.600	
Fraction refinancing		0.040	
Fixed rate to fixed rate refinancing		0.701	
Fraction with negative ADL incentive		0.563	
Of which refinance		0.013	
Fraction with positive ADL incentive		0.437	
Of which refinance		0.076	
	All	At higher incentives	At lower incentives
Panel B. Household observations			
Number of households	614,811		
Fraction never refinancing	0.498		
Fraction refinancing once	0.402	0.276	0.126
Fraction refinancing twice	0.089	0.040	0.049
Fraction refinancing three or more times	0.011	0.002	0.009

Notes: These statistics are calculated using all unique mortgages taken by households in Denmark with an unchanging number of adult members, and with a single fixed rate mortgage. In panel A, the rows show, in order, the total number of household-quarter observations in the data; the average principal remaining (outstanding) on these mortgages in millions of Danish kroner (DKK); the average number of years remaining before mortgages mature; the average loan-to-value (LTV) ratio on these mortgages; the fraction refinancing in a quarter (i.e., the fraction of households that refinanced their preexisting mortgage voluntarily rather than refinancing for exogenous reasons such as moving house); the fraction refinancing from a fixed rate mortgage (FRM) to another FRM; the fraction of all mortgages with negative refinancing incentives calculated using the Agarwal, Driscoll, and Laibson (2013) formula; the fraction of mortgages with negative ADL incentives which refinance; the fraction of mortgages with positive refinancing incentives computed using the ADL formula; and the fraction with positive ADL incentives which refinance. In panel B, the rows document the fractions of households refinancing either once or multiple times during the sample period. The columns document the fractions of these households refinancing at incentives which are either greater than, or less than or equal to, incentive levels previously experienced over the sample period.

- The key behavioral fact is that only 7.6 percent of positive-incentive household-quarters refinance, while 1.3 percent of negative-incentive household-quarters refinance.
- This means positive incentives matter, but they are far from sufficient to produce prompt action.

5.2 Figures 2 and 3: slow refinancing despite positive incentives

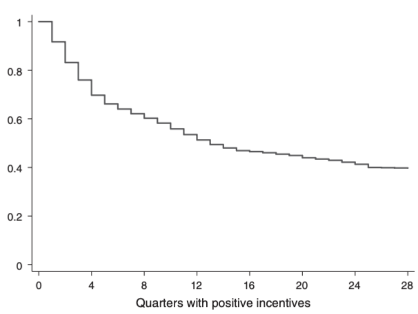
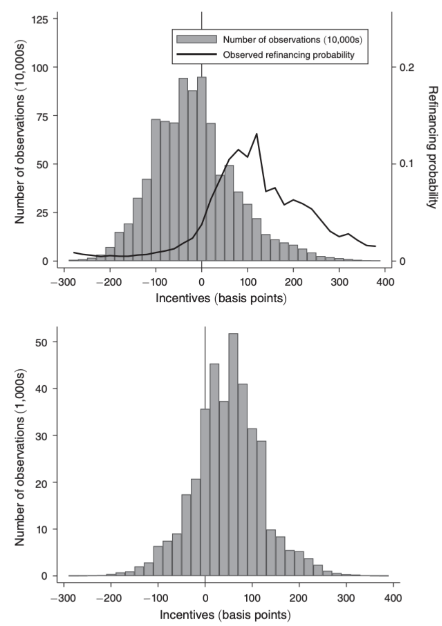


FIGURE 2. EVALUATING REFINANCING ACTIVITY RELATIVE TO THE ADL THRESHOLD (*continued*)

Notes: This figure illustrates refinancing activity in the sample evaluated against the household-quarter-specific ADL threshold. The top plot shows the histogram of computed incentives with the refinancing probability superimposed on it; the second plot shows the number of refinancings at each point corresponding to the dark line on the top plot; and the third plot shows the Kaplan-Meier "survival" (i.e., non-refinancing) estimate, i.e., plotting the number of quarters at which the household has positive incentives but does not refinance, accounting for data censoring.



(*Continued*)

FIGURE 2. EVALUATING REFINANCING ACTIVITY RELATIVE TO THE ADL THRESHOLD

Read:

- Figure 2 shows that many households have positive refinancing incentives, but refinancing probabilities remain low.

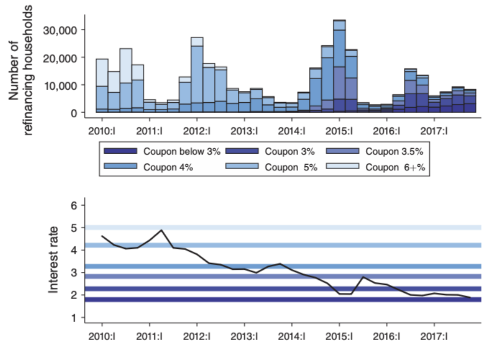


Figure 3. Refinancing Activity by Preexisting Mortgage Coupon Rates

Notes: This figure illustrates the history of refinancing activity in the sample of Danish fixed-rate mortgages. In the top plot, the bars represent the number of refinancing households in each quarter. The bars are shaded according to the coupon rate on the old mortgage from which households refinance. In the bottom plot, we show the evolution of the quarterly Danish mortgage interest rate as it moves through the average refinancing threshold for mortgages with a specific coupon rate. For example, the very top lightest shaded horizontal line in the bottom plot shows the average interest rate refinancing threshold for the group of mortgages that bear coupon rates of 6 percent, i.e., the point at which the current interest rate needs to be, on average, to optimally justify refinancing for this group of mortgage holders. The thresholds for mortgages with coupons of 6 percent, 5 percent, 4 percent, 3.5 percent, 3 percent, and 2.5 percent are 5.06 percent, 4.22 percent, 3.27 percent, 2.83 percent, 2.28 percent, and 1.83 percent respectively.

- About 18 percent of refinancings occur at negative incentives, while many large positive incentives are ignored.
- The Kaplan–Meier survival curve shows that even four years after a positive incentive appears, roughly one-half of mortgages still have not refinanced.
- Figure 3 shows several refinancing waves: 2010, 2012, 2014–2015, and a smaller wave in 2016–2017.
- These waves occur as mortgage rates decline through thresholds for different old coupon groups.
- The figure makes the institutional mechanism clear: lower rates create refinancing opportunities, but households respond slowly and in waves rather than instantly.

5.3 Figure 4: rejecting a fixed household-specific threshold

Read:

- The top panel compares the incentive at the refinancing date with the best earlier incentive.
- About 35 percent of refinancing households refinance at an incentive lower than an incentive they had already ignored.
- The middle panel focuses on households that refinance at least twice. The second refinancing often occurs at a very different incentive from the first one.
- The standard deviation of the difference between the second and first refinancing incentives is about 327 basis points.
- The bottom panel shows that households often wait several years even after returning to an incentive above a previously revealed threshold.

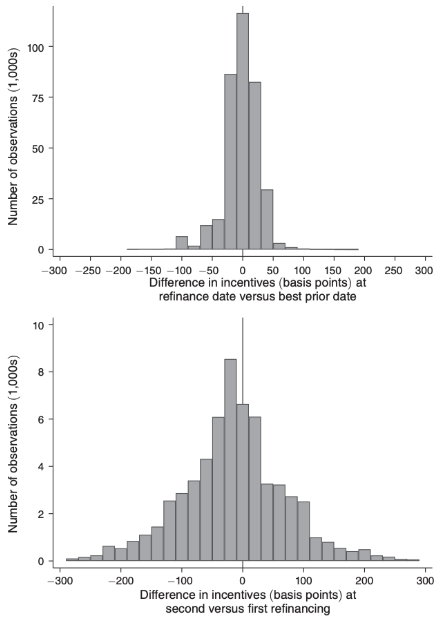


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FIGURE 4. EVALUATING WHETHER REFINANCING ACTIVITY OCCURS AT A HOUSEHOLD-SPECIFIC THRESHOLD

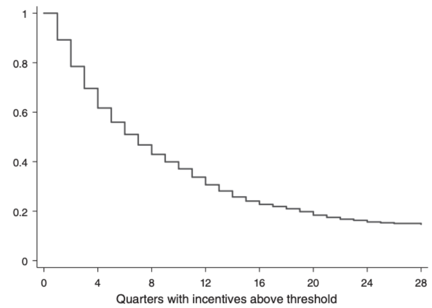

FIGURE 4. EVALUATING WHETHER REFINANCING ACTIVITY OCCURS AT A HOUSEHOLD-SPECIFIC THRESHOLD
(*continued*)

Notes: This figure illustrates refinancing activity to check whether households have idiosyncratic thresholds at which they always refinance. The top plot shows the histogram of the difference between computed incentives at the point at which households refinance and the best ever incentive experienced over the sample period prior to the point of refinancing; the second plot focuses on households that refinance at least twice, and plots the histogram across these households of the difference between the incentives at the two refinancing points; and the third plot shows the Kaplan-Meier "survival" (i.e., non-refinancing) estimate, i.e., plotting the number of quarters at which the household has incentives above those which they previously refinanced at but do not subsequently refinance, accounting for data censoring.

- Main message: a pure state-dependent fixed-threshold model is not enough.

5.4 Figure 5: refinancing efficiency and missed savings

Read:

- Refinancing efficiency is the ratio of realized interest savings to the savings from the optimal ADL strategy.
- Average realized savings are about 55 basis points of mortgage principal, while optimal savings are about 98 basis points.
- The implied missed savings average about 43 basis points of principal, or roughly DKK 2,700 per year and 58 basis points of household income.
- Efficiency is hump-shaped in age: it peaks near 75 percent around the twenty-fifth percentile of age, then falls for older households to about 60 percent.
- Efficiency increases with education, income, and housing wealth, from roughly 50 percent to about 75 percent.
- Financial wealth has a much flatter relationship with realized refinancing efficiency.

5.5 Table 2: comparing choice models

Read:

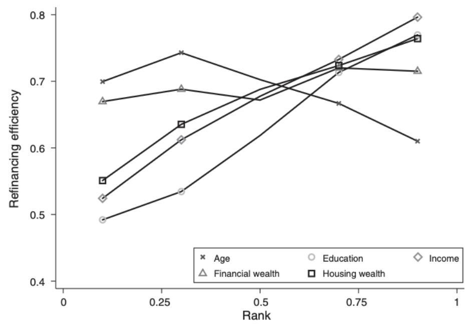

FIGURE 5. REFINANCING EFFICIENCY

Notes: This figure plots the average refinancing efficiency, calculated as the ratio of actual savings to counterfactual savings (counterfactual estimated under optimal refinancing), as a function of the ranked variables of age, education, income, financial wealth, and housing wealth.

TABLE 2—CHOICE MODELS

	Model 1	Model 2	Model 3	Model 4	Model 5
β	−1.350	−1.063	1.298	0.745	0.846
φ_m		6.126		2.437	3.289
Current quarter dummies		No		No	No
Mortgage age dummies		No		No	No
Demographics		No		No	No
φ_{sd}					1.300
χ			2.381	1.671	0.967
Current quarter dummies			No	No	No
Mortgage age dummies			No	No	No
Demographics			No	No	No
Pseudo R^2	−3.508	−0.096	−0.019	0.000	0.007

	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11
β	0.655	0.667	0.677	0.770	0.820	0.873
φ_m	3.149	3.785	2.465	2.219	2.547	2.580
Current quarter dummies	Yes	Yes	No	No	No	No
Mortgage age dummies	No	Yes	No	No	No	No
Demographics	No	No	No	No	Yes	Yes
φ_{sd}						0.848
χ	1.387	1.392	2.238	3.049	3.059	2.988
Current quarter dummies	No	No	Yes	Yes	Yes	Yes
Mortgage age dummies	No	No	No	Yes	Yes	Yes
Demographics	No	No	No	No	Yes	Yes
Pseudo R^2	0.029	0.035	0.041	0.052	0.069	0.070
Unconstrained out-of-sample pseudo R^2 average (2012–2017)	0.001	0.008	0.013	0.020	0.037	
Unconstrained out-of-sample average (2012–2017)	0.039	0.039	0.055	0.059	0.067	

Notes: We estimate our models using all households in Denmark with an unchanging number of household members, with a single fixed rate mortgage in the beginning of each year from 2010–2017. The dependent variable takes the value of 1 if there is refinancing in a given quarter, and 0 otherwise. Each column lists the parameters of our model of refinancing: χ is the probability that a household is asleep and does not respond to refinancing incentives (i.e., probability $= \exp(\chi)/(1+\exp(\chi))$). φ_m captures the average level of psychological refinancing costs, and depending on the model variation listed in the rows, χ and φ_m can be functions of demographic characteristics, and χ can also depend on mortgage age effects and current quarter effects. φ_{sd}, when included, estimates a random coefficient model for psychological refinancing costs. $\exp(\beta)$ captures responsiveness to incentives. Pseudo R^2 is calculated using the formula $R^2 = 1 - \mathcal{L}_1/\mathcal{L}_0$, where $\mathcal{L}_1$ is the log-likelihood from the model specified in the column header, and $\mathcal{L}_0$ is the log-likelihood from a model which only allows for a constant probability of being asleep, a stochastic choice error, and a constant fixed psychological refinancing cost (i.e., Model 4). Out-of-sample pseudo R^2 for Models 6–10 are estimated in expanding windows 2010–2011, 2010–2013, and 2010–2015, and use the resulting estimates to forecast refinancing in three two-year windows 2012–2013, 2014–2015, and 2016–2017. We use two approaches: the first approach sets the time effects to the average value estimated over the in-sample expanding window. The second approach allows time fixed effects to be freely estimated in the out-of-sample windows. All coefficients in the table are significant at the 1 percent level or less.

- Table 2 compares increasingly rich models of refinancing behavior.
- The simplest incentive-only model fits very poorly.
- A model with only psychological costs fits better but implies implausibly high costs.
- A model with only asleep probabilities also improves fit, but the best fit comes from combining state-dependent and time-dependent inaction.
- Model 10 is the baseline: it includes demographics in both psychological costs and asleep probabilities, and includes time and mortgage-age effects in the asleep probability.
- Model 10 reaches pseudo-$R^2 = 0.069$.
- Adding random coefficients for unobserved heterogeneity in psychological costs raises pseudo-R^2 only to 0.070, so the added complexity is not useful.

5.6 Figures 6, 7, and 8: model fit, cost components, and asleep probabilities

Read:

- Figure 6 shows that the baseline model tracks the observed refinancing-probability curve reasonably well.

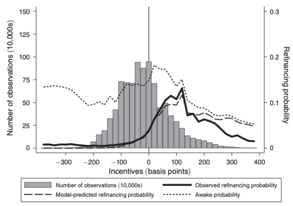

FIGURE 6. REFINANCING, INCENTIVES, AND MODEL IMPLIED REFINANCING PROBABILITIES

Notes: This figure plots refinancing probabilities from the baseline model presented in Table 3, as a function of refinancing incentives, alongside the number of observations at each level of incentives. The bars in this figure show the number of household-quarters (scale on the left vertical axis) and the lines show the fraction of these household-quarters that refinance (scale on the right vertical axis), both plotted for each level of refinancing incentives shown on the horizontal axis. The bars are 20-basis-point incentive intervals centered at the points on the horizontal axis. The solid line shows the actual refinancing probability observed in the data, the long-dashed line shows the model-predicted refinancing probability, and the short-dashed line shows the fraction of households that do not perceive incentives as estimated by the model (i.e., time-dependent inaction) in each period.

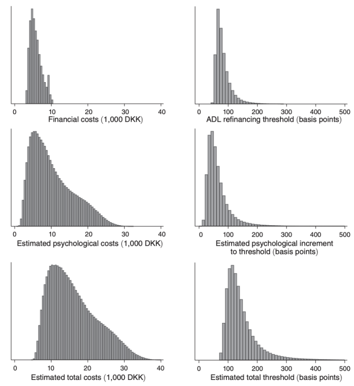

FIGURE 7. REFINANCING COST COMPONENTS

Notes: These figures summarize the costs of refinancing estimated from the baseline model presented in Table 3 over the entire sample period. The three plots in the left column show the costs in 1,000 DKK, while the three plots in the right column show these costs in the form of the implied interest rate threshold in basis points that they translate into using the ADL (2013) function. Descending vertically, the first row shows the pure financial costs of refinancing, which are based on mortgage size. The second row shows the estimated psychological costs of refinancing, while the third row is the total costs, which sum the two rows above it.

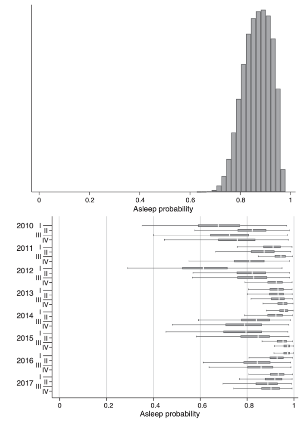

FIGURE 8. MODEL IMPLIED TIME-DEPENDENT INACTION PROBABILITY

Notes: This figure shows the model implied probability of households being subject to time-dependent inaction, estimated using the model in Table 3. The top panel shows a histogram of the distribution of the estimated time-dependent inaction probability across households, computed using a representative quarter, i.e., inputting the average mortgage age effect and average current quarter time effect estimated in the data. The bottom panel shows a box plot of the model implied estimated time-dependent inaction probability for each quarter of our data, i.e., inputting the time effect and mortgage age effect for each quarter listed on the vertical axis.

- The model decomposes the curve into an incentive-response part and an awake/asleep probability part.
- Figure 7 decomposes refinancing costs.
- Direct financial costs average about **DKK 5,850**.
- The ADL threshold averages about **83 basis points**, with standard deviation about **37 basis points**.
- Psychological costs average about **DKK 10,400**.
- Total refinancing thresholds average about **146 basis points**, with standard deviation about **60 basis points**.
- Figure 8 shows that the average probability of being asleep is about **87 percent**, with substantial time variation.
- High refinancing-wave quarters are interpreted as quarters when more households are awake or actively considering refinancing.

5.7 Table 3 and Figure 9: who has which friction?

Read:

- Table 3 estimates demographic effects on both psychological costs and asleep probabilities.
- Financial literacy in the household or extended family is associated with faster refinancing.
- Life events such as getting married or having children are also associated with more active refinancing, plausibly because they prompt broader financial reconsideration.
- Figure 9 is the central cross-sectional mechanism figure.

Table 3—Model with Heterogeneous Household Characteristics

	β	φ	χ
Intercept	0.820 (0.004)	2.547 (0.026)	3.059 (0.026)
Random coefficient component			
Single male household		−0.088 (0.025)	0.008 (0.017)
Single female household		−0.106 (0.027)	−0.107 (0.018)
Married household		0.101 (0.016)	−0.039 (0.011)
Children in family		0.112 (0.015)	0.101 (0.011)
Immigrant		−0.101 (0.019)	0.163 (0.013)
Financially literate		−0.160 (0.023)	−0.018 (0.019)
Family financially literate		−0.001 (0.014)	−0.093 (0.011)
Getting married		−0.248 (0.055)	−0.070 (0.031)
Having children		−0.106 (0.025)	−0.087 (0.016)
Region of Northern Jutland		0.112 (0.018)	−0.280 (0.014)
Region of Middle Jutland		0.082 (0.015)	−0.215 (0.011)
Region of Southern Denmark		0.024 (0.016)	−0.100 (0.012)
Region of Zealand		0.064 (0.017)	0.142 (0.012)
Demeaned rank of:			
Age		−0.094 (0.035)	0.783 (0.022)
Length of education		0.110 (0.021)	−0.251 (0.015)
Income		0.815 (0.035)	−0.763 (0.023)
Financial wealth		0.906 (0.025)	−0.242 (0.017)
Housing wealth		0.636 (0.025)	−0.814 (0.017)
Nonlinear transformation $f(x)$, x is the demeaned rank of:			
Age		−1.323 (0.064)	−0.014 (0.037)
Length of education		0.275 (0.038)	−0.005 (0.027)
Income		−0.401 (0.043)	0.615 (0.029)
Financial wealth		−0.896 (0.039)	0.150 (0.027)
Housing wealth		−0.546 (0.039)	0.380 (0.027)
Current quarter dummies		Yes	
Mortgage age dummies		Yes	
Pseudo R^2		0.069	
log-likelihood		−1,332,195	
Observations		9,351,183	

Notes: We estimate this specification using all households in Denmark with an unchanging number of household members, with a single fixed rate mortgage in the beginning of each year from 2010–2017. The dependent variable takes the value of 1 for a refinancing in a given quarter, and 0 otherwise. Each column lists the parameters of our model of refinancing. $\exp(\beta)$, which does not depend on demographics, captures the responsiveness to incentives. χ captures the probability that a household is asleep and does not respond to refinancing incentives, and the rows show its dependence on the listed demographic characteristics. And φ captures the level of psychological refinancing costs once again as a function of demographic characteristics. The coefficients include nonlinear transformations, $f(x)$, of all the ranked control variables in addition to their levels, where $f(x) = \sqrt{2}x^2$. Pseudo R^2 is calculated using the formula $R^2 = 1 - \mathcal{L}_1/\mathcal{L}_0$, where $\mathcal{L}_1$ is the log-likelihood from the given model and $\mathcal{L}_0$ is the log-likelihood from a model which only allows for a constant probability of being asleep. Standard errors clustered at the level of households are reported in parentheses.

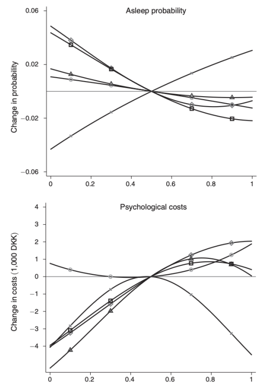

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Figure 9. Marginal Effects of Ranked Variables

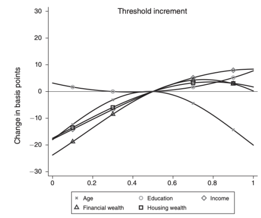

Figure 9. Marginal Effects of Ranked Variables (*continued*)

Notes: This figure shows the marginal change in the probability of being subject to time-dependent inaction, the estimated psychological costs of refinancing in 1,000 DKK, and the psychological cost increment to the interest-rate threshold to be surmounted to induce a household to refinance, all as functions of selected ranked variables: age, education, income, financial wealth, and housing wealth. To plot these marginal effects, we use the household-level fitted values of the baseline model presented in Table 3.

- Older households are more likely to be asleep.
- Households with higher education, income, financial wealth, and housing wealth are less likely to be asleep.
- Psychological costs in DKK are highest for middle-aged households and for households with higher socioeconomic status.
- Therefore, older and poorer households mainly suffer from time-dependent inaction, while wealthier middle-aged households mainly look like they have high state-dependent costs.

5.8 Robustness exercises

Read:

- The main results survive several sample restrictions, including removing cash-out refinancing and FRM-to-ARM refinancing.
- The results are also robust to focusing only on larger mortgages with long remaining maturity, where threshold calculations are less sensitive to assumptions.
- Alternative parameter choices for the ADL model do not materially change the conclusions.
- Replacing ADL with the Chen–Ling optimal refinancing model shifts some thresholds but leaves the main behavioral patterns intact.
- Prompt refinancers provide an intuitive benchmark: their behavior is close to the ADL threshold, which supports using ADL as the rational benchmark.

5.9 Figure 10: policy experiments and monetary transmission

Read:

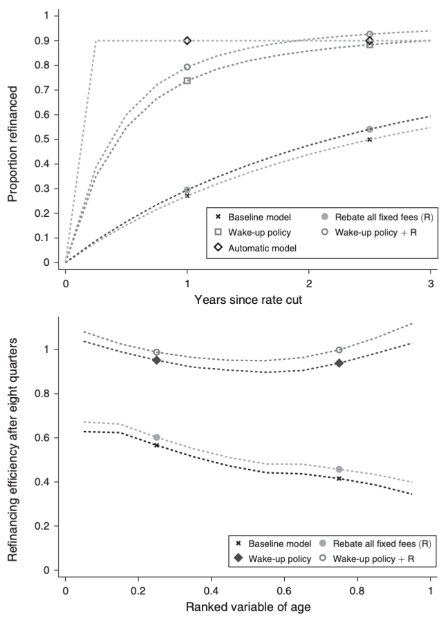

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Figure 10. Policy Experiments

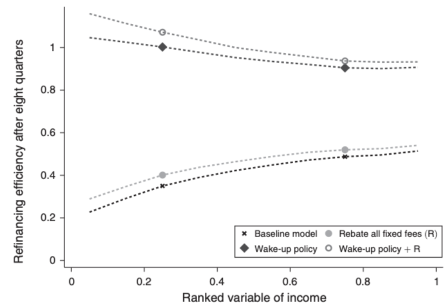

Figure 10. Policy Experiments (*continued*)

Notes: These figures consider policies to induce household refinancing alongside an interest rate cut in which 90 percent of Danish households have a refinancing incentive exceeding their ADL (2013) threshold: a policy in which mortgages automatically refinance when the interest rate saving exceeds the ADL threshold; a policy that "wakes up" households, cutting the asleep probability in half from its initial level; a policy that rebates all fixed fees incurred by households; a policy that combines "waking up" with the rebate; and a "do nothing policy" in which households refinance according to our baseline model. The top panel shows the fraction refinancing at each point in time, and the second (third) the fraction refinancing 8 quarters post-cut along the age (income) distribution.

- Figure 10 studies policies after an interest-rate cut that gives 90 percent of households positive ADL incentives.
- Fully rational households would refinance immediately up to about 90 percent.
- A pure state-dependent model with psychological costs generates an immediate but smaller refinancing wave of just above 60 percent.
- Time-dependent inaction produces a gradual response because households wake up only over time.
- A fee rebate helps, especially for smaller mortgages, but it is less powerful than a wake-up policy.
- The wake-up policy, modeled as cutting asleep probabilities in half, is the most effective way to raise refinancing by older and lower-income households.
- Main policy lesson: the refinancing channel of monetary policy depends not only on interest-rate reductions and fees, but also on households' attention to opportunities.

6 Short summary

- **Method.** The paper estimates a mixture model of mortgage refinancing that combines state-dependent threshold behavior with time-dependent inattention.
- **Data.** It uses Danish administrative data from 2009–2017, covering more than 9.3 million household-quarter observations of fixed-rate mortgage borrowers.

- **Identification.** Denmark's refinancing system removes many standard credit and collateral constraints, while panel data reject a pure fixed-threshold explanation.
- **Main result.** Middle-aged and wealthy households behave as if they face high psychological refinancing costs, while older, poorer, and less-educated households are often asleep and refinance with low probability regardless of incentives.
- **Policy result.** Policies that make households aware of refinancing opportunities can improve refinancing efficiency more than fee rebates, especially for vulnerable demographic groups.

7 Conclusion

7.1 Core conclusion

- The paper shows that slow mortgage refinancing is not explained by a single friction.
- A pure state-dependent model with fixed household-specific thresholds is rejected by panel behavior.
- A model with both psychological costs and time-dependent inaction fits the data and explains meaningful cross-sectional heterogeneity.
- The decomposition matters for policy because different frictions call for different interventions.

7.2 Economic intuition

- If the main problem is a high threshold, lowering fees or increasing interest savings can induce refinancing.
- If the main problem is that households are asleep, stronger incentives alone may not generate prompt action.
- The model finds that wealthier middle-aged households look more threshold-constrained, while older and poorer households look more attention-constrained.
- This distinction explains why slow refinancing both reduces household savings and weakens the pass-through of expansionary monetary policy.

7.3 Takeaway

This paper is important because it turns a common household-finance puzzle into a measurable decomposition. Households do not merely fail to refinance because fees are high or because refinancing is complicated. Many households appear not to consider refinancing opportunities at all in a given period. For fixed-rate mortgage economies, this means that the effectiveness of lower interest rates depends partly on information, attention, and household financial capability, not only on market rates or mortgage contract design.